

GOLI CHIRANJEEVI

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with experience in Python, engineering data analysis, and automation using Python, CAPL, and Vector CANoe through a Graduate Apprenticeship at Volvo Group. Built AI-powered projects including a RAG knowledge assistant and a natural language-to-SQL application. Passionate about applying data and AI to solve real-world problems through continuous learning and collaboration.

TECHNICAL SKILLS

Programming & Data Analysis: Python, SQL, Pandas, NumPy, Matplotlib, Statistical Analysis, C, CAPL
AI & Machine Learning: LLMs, Retrieval-Augmented Generation (RAG), Semantic Search, Vector Embeddings, Prompt Engineering
Databases & Backend Development: FastAPI, MySQL, SQLAlchemy, REST APIs
Engineering Domain Knowledge: CAN, LIN, TCP/IP, DBC/LDF Configuration, ECUs
Tools & Platforms: Git, GitHub, VS Code, Postman, Excel, Vector CANoe, Jira, ET tool
AI Tools: Claude, ChatGPT, Google AI Studio
Soft Skills: Adaptability, Attention to Detail, Problem Solving, Ownership & Accountability, Learn-by-Doing

WORK EXPERIENCE

Graduate Apprentice Trainee – Data Automation & Test Engineering

May 2025 – May 2026

Volvo Group India Pvt. Ltd.

Data Pipeline Development & Test Automation

- Built Python utilities to parse Excel-based test specifications, perform engineering unit conversions, and auto-generate CAPL function calls, reducing manual scripting time by 90%.
- Engineered Excel-driven automation scripts for multi-channel test execution, reducing manual test setup time by 90% while improving scalability across parallel test runs.
- Designed and configured Vector CANoe test environments integrating T2 and T3 hardware platforms, supporting reliable, repeatable laboratory validation.
- Developed CAPL scripts and control panels for signal access and automated logging across thermal chambers, alternators, and DC regulated power supplies, streamlining data capture for downstream analysis.
- Delivered CAPL-based automation supporting PMSL test execution alongside CANoe/CAPL and Python, improving repeatability and reducing manual intervention across multiple test scenarios.

Data Acquisition, Network Communication & Hardware Configuration

- Maintained CAN network configuration data by creating and correcting DBC files and CAN matrix content, with every change validated through structured technical review.
- Configured and validated LIN communication for alternator networks, including LDF setup and wiring, and verified combined CAN/LIN interaction scenarios.
- Independently researched and implemented TCP/IP socket communication in Vector CANoe using CAPL, enabling integration with an external DC regulated power supply for automated testing.
- Developed a Python automation tool that converted Excel-based test configurations into Neware XML files, eliminating manual channel setup and enabling simultaneous execution across 8 test channels.
- Flashed target software onto T2 and T3 ECU variants to prepare consistent and reproducible test environments for validation.

Lab Operations & Documentation

- Operated and maintained laboratory equipment – multimeters, oscilloscopes, programmable power supplies, and electronic loads – to collect accurate performance data for analysis.

- Prepared technical documentation for developed automation tools and maintained lab equipment checklists to track operational status and readiness.

PROJECTS

FleetMind AI – RAG-Based Knowledge Agent for Heavy-Duty Truck Maintenance

Python, RAG, ChromaDB, Gemini API

- Designed and built an interactive Retrieval-Augmented Generation (RAG) assistant with Python, Streamlit, ChromaDB, Sentence Transformers, and the Google Gemini API to support fleet maintenance troubleshooting and predictive-maintenance decisions over heavy-duty vehicle manuals.
- Engineered an agentic retrieval-routing layer that evaluated retrieval confidence and triggered clarification prompts for ambiguous queries, improving answer reliability and reducing incorrect responses.
- Built a document ingestion pipeline with PyMuPDF and LangChain that extracted, chunked, embedded, and indexed 1,212 pages into 3,795 semantic chunks, enabling fast semantic search over technical fleet documentation.

AI Data Analyst Agent

Python, Streamlit, Gemini API, SQLAlchemy, MySQL

- Built an interactive data analyst assistant with Streamlit and the Google Gemini API that lets users upload any CSV dataset and ask analytical questions in plain English.
- Implemented automated schema detection and natural-language-to-SQL generation, executing queries against a MySQL database via SQLAlchemy and returning clear, plain-text answers.
- Added automatic chart generation for trend and comparison questions plus context-aware follow-up handling, so it retains prior questions within a session.

Asset Management System

Python, FastAPI, MySQL, SQLAlchemy

- Built a REST API-based Employee Management System with FastAPI, SQLAlchemy, and MySQL, supporting complete CRUD operations end to end.
- Designed database models and implemented request validation with Pydantic, using dependency-based database session management for reliable, testable endpoints.

EDUCATION

B.Tech, Electronics and Communication Engineering

Vasireddy Venkatadri Institute of Technology, Guntur

2020 – 2024

CGPA: 8.92

CERTIFICATIONS

- **AI For Everyone** – DeepLearning.AI, Coursera
- **Python Data Structures** – University of Michigan, Coursera